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Adaptive fitting with THB-splines: Error analysis and industrial applications

Cesare Bracco [a], Carlotta Giannelli [a,*], David Großmann [b], Alessandra Sestini [a]

[a] Dipartimento di Matematica e Informatica "U. Dini", Università degli Studi di Firenze, Italy
[b] MTU Aero Engines AG, Munich, Germany

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A B S T R A C T

Automatic fitting techniques are required in many industrial applications, as for example, instrument calibration, data analysis, geometric modeling or reverse engineering. We present an error analysis of advanced techniques for scattered data approximation based on truncated hierarchical B-spline (THB-spline) constructions. The adaptive fitting framework is validated on data of industrial complexity related to geometric models of different aircraft engine parts. We demonstrate that the twofold adaptive nature of the method, which takes into account both the *shape* and the *distribution* of the data, is crucial to define an effective fully automatic scheme with suitable local refinement capabilities.

1. Introduction

Automatic fitting techniques that guarantee high precision, while simultaneously providing shapes free from artifacts, have been widely investigated over the years. In industrial software environments where commercial Computer Aided Design (CAD) systems are commonly used, the need of overcoming the limits of well established CAD procedures is currently emerging. New solutions in this direction will allow the development of more flexible surface reconstruction (and modeling) schemes. For example, even if the standard B-spline model provides accurate approximations exploiting efficient and numerically stable evaluation and manipulation algorithms, the lack of local refinement of the tensor-product construction, together with the restriction to rectangular topologies, demands alternative solutions which outperform the schemes based on commercial standards. This is particularly evident in the surface reconstruction process of unevenly distributed data arising in different contexts, that may vary from instrument calibration and data analysis, to geometric modeling and reverse engineering. In all these cases, a highly varying density of the data distribution does not nicely fit with uniform tensor-product geometries. Fig. 1 shows two challenging parts of the aircraft engine industry, a blade and a tensile, where we will concentrate on the most critical regions of their scanned data, shown in Fig. 2. Scanned data sets, generated by an optical measurement system, are heavily used in todays engineering workflows to analyze mechanical parts of real world shapes.

Everyday practice in the development of approximation and fitting schemes of challenging industrial components suggests that unwanted oscillations in the reconstructed spline model may partially destroy the quality of the CAD geometry. The common solutions currently available to overcome this problem, as well as related issues, mainly rely on suitable selections of different shape parameters, connected to the knot configurations, the smoothing parameters, as well as optimal parameterization construction. In particular two types of different oscillations may arise when too few or too many degrees

* Corresponding author.
E-mail addresses: cesare.bracco@unifi.it (C. Bracco), carlotta.giannelli@unifi.it (C. Giannelli).

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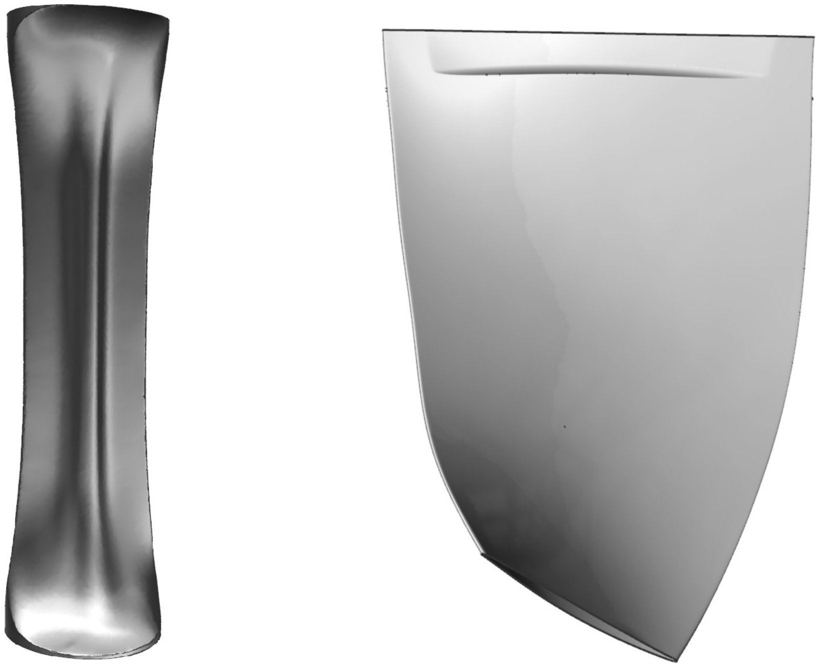


Fig. 1. Two different aircraft engine parts: a tensile (left) and a blade (right) model.

of freedom are (locally) available. In the first case, the geometry oscillates *smoothly* in transition from low-to-high curvature parts and local refinement schemes may be used to address this problem by adding more degrees of freedom only in the area of interest. On the other side, when too many degrees of freedom (with respect to the local number of data) are available in a certain area of the domain, the reconstructed geometry may exhibit oscillations characterized by small *sharp* peaks. In this case, a proper handling of the scattered data configuration is crucial to provide an advanced fitting scheme which automatically captures the regions of high and low data density.

Local refinement schemes with adaptive splines defined on extensions of tensor-product meshes which allow T-junctions were investigated by several authors, leading to different spline constructions. We mention here hierarchical B-splines (Forsey and Bartels, 1988; Kraft, 1997), T-splines (Sederberg et al., 2003), PHT-splines (Deng et al., 2006, 2008), THB-splines (Giannelli et al., 2012), and LR-splines (Dokken et al., 2013). Related surface fitting schemes include Forsey and Bartels (1995), Greiner and Hormann (1997), Deng et al. (2008), Wang et al. (2011) and more recently Kiss et al. (2014), Skytt et al. (2015, 2017). Model simplification in terms of data reduction schemes based on different spline constructions were also considered, see e.g., Sederberg et al. (2004), Bracco et al. (2017b).

An efficient solution to properly address a non-uniform data distribution, which also avoids the solution of a global linear system, relies in the so-called *two-stage methods* which combine local approximations (first stage) with suitable spline constructions (second stage). In particular, scattered data fitting by direct extension of local polynomials to bivariate splines was originally introduced in Davydov and Zeilfelder (2004) and subsequently extended in Davydov et al. (2005, 2006, 2014). An adaptive extension of local least squares approximations to hierarchical spline spaces was also recently proposed (Bracco et al., 2017a). Alternative constructions include the possibility of considering variable degree splines, see e.g. Costantini (2000), Sederberg et al. (2003), Shen et al. (2013).

In this paper, we combine and extend recent results obtained with truncated hierarchical B-splines for the design of adaptive approximation schemes suitable for industrial applications. While the local refinement capabilities of THB-splines were already exploited for efficient adaptive CAD model reconstruction by considering a *global* regularized least square fitting (Kiss et al., 2014), the varying density of the data distribution was not yet taken into account. By following Bracco et al. (2017a), we address this point by exploiting *local* polynomial approximations of variable degree in connection with hierarchical quasi-interpolation. We enhance the flexibility of scattered data approximation with THB-splines by designing and analyzing an algorithm that does not depend on the polynomial basis associated to the local approximations. In addition, we develop the error analysis of the method, as well as a refinement strategy that allows us to consider a varying number of local data points. This strategy reduces oscillations due to local overfitting.

The remainder of the paper consists of four sections. The (local) adaptive fitting scheme and its error analysis are presented in Section 2 and 3, respectively. The application of the scheme to the reconstruction of critical CAD geometries is presented in Section 4, while Section 5 concludes the paper. The notation considered in the paper is summarized in the Appendix.

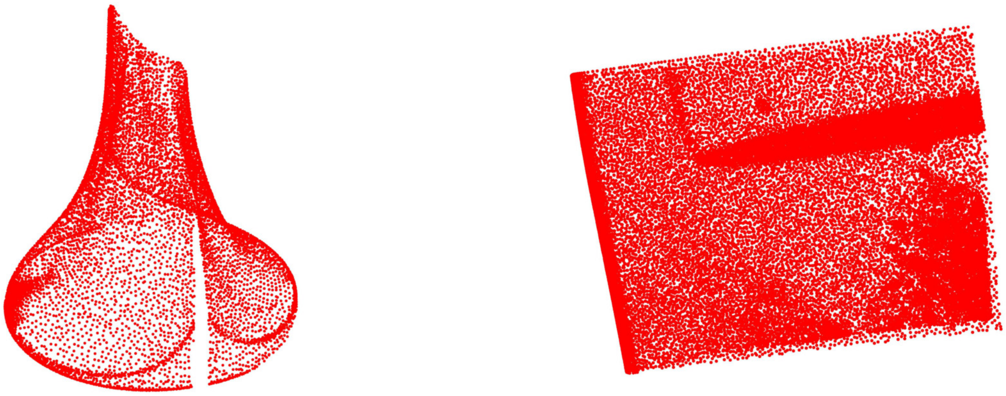


Fig. 2. The scanned data points corresponding to critical regions of the tensile (left) and the blade (right) shown in Fig. 1.

2. Two-stage approximation with THB-splines

In this section, after briefly recalling the definition and main properties of truncated hierarchical B-splines (THB-splines), we describe the adaptive two-stage approximation method for scattered data fitting which combines local polynomial approximations (first stage) with hierarchical quasi-interpolation (second stage).

2.1. THB-splines

Let us consider a sequence of nested bivariate tensor-product spline spaces of degree $\mathbf{d} := (d_1, d_2)$ with their corresponding tensor-product grids

$$V^0 \subset V^1 \subset \ldots \subset V^{M-1}, \qquad \mathcal{G}^0, \mathcal{G}^1, ..., \mathcal{G}^{M-1},$$

defined on a closed domain $\Omega \subset \mathbb{R}^2$. For $\ell = 0, ..., M-1$, we denote by

$$\mathcal{B}^\ell_{\mathbf{d}} := \left\{ B^\ell_J,\ J \in \Gamma^\ell_{\mathbf{d}} \right\}$$

the B-spline basis of V^ℓ, where $\Gamma^\ell_{\mathbf{d}}$ is a suitable set of multi-indices. Moreover, let

$$\Omega^0 \supseteq \Omega^1 \supseteq \ldots \supseteq \Omega^M,$$

be a nested sequence of closed domains, where each $\Omega^\ell \subset \Omega$ is composed of cells belonging to $\mathcal{G}^\ell$, for $\ell = 0, \ldots, M-1$, and $\Omega^0 = \Omega$, $\Omega^M = \emptyset$. The hierarchical mesh $\mathcal{G}_\mathcal{H}$ is the set of all active cells belonging to $\Omega^\ell \setminus \Omega^{\ell+1}$, for each level $\ell = 0, \ldots, M-1$.

The hierarchical B-spline basis is defined according to the following construction which collects the set of active basis functions at different levels, analogously to the iterative definition introduced in Vuong et al. (2011).

Definition 1. The hierarchical B-spline $\mathcal{H}_{\mathbf{d}}(\mathcal{G}_\mathcal{H})$ basis of degree $\mathbf{d}$ with respect to the mesh $\mathcal{G}_\mathcal{H}$ is

$$\mathcal{H}_{\mathbf{d}}(\mathcal{G}_\mathcal{H}) := \left\{ B^\ell_J \in \mathcal{B}^\ell_{\mathbf{d}} \ :\ J \in A^\ell_{\mathbf{d}},\ \ell = 0, ..., M-1 \right\},$$

where

$$A^\ell_{\mathbf{d}} := \{ J \in \Gamma^\ell_{\mathbf{d}} : \operatorname{supp} B^\ell_J \subseteq \Omega^\ell \ \wedge\ \operatorname{supp} B^\ell_J \not\subseteq \Omega^{\ell+1} \}$$

is the active set of multi-indices $A^\ell_{\mathbf{d}} \subseteq \Gamma^\ell_{\mathbf{d}}$ and $\operatorname{supp} B^\ell_J$ denotes the intersection of the support of B^ℓ_J with Ω^0.

We denote by $S_\mathcal{H} := \operatorname{span} \mathcal{H}_{\mathbf{d}}(\mathcal{G}_\mathcal{H})$ the space spanned by the hierarchical B-splines and associated to the mesh $\mathcal{G}_\mathcal{H}$. Since they are linearly independent, the hierarchical B-splines are a basis of $S_\mathcal{H}$.

We introduce the truncation operator $\operatorname{trunc}^{\ell+1} : V^\ell \to V^{\ell+1}$ as follows. For any $s \in V^\ell$ with representation in the B-spline basis of $V^{\ell+1}$ given by

$$s = \sum_{J \in \Gamma^{\ell+1}_{\mathbf{d}}} \sigma^{\ell+1}_J B^{\ell+1}_J,$$

the truncation of s with respect to level $\ell+1$ is defined as

$$\mathrm{trunc}^{\ell+1}(s) := \sum_{J \in \Gamma_{\mathbf{d}}^{\ell+1} \,:\, \mathrm{supp}\, B_J^{\ell+1} \not\subseteq \Omega^{\ell+1}} \sigma_J^{\ell+1} B_J^{\ell+1}.$$

The (cumulative) truncation operator $\mathrm{Trunc}^{\ell+1} : V^{\ell} \to S_{\mathcal{H}} \subseteq V^{M-1}$ with respect to all finer levels in the hierarchy is then defined as

$$\mathrm{Trunc}^{\ell+1}(s) := \mathrm{trunc}^{M-1}(\mathrm{trunc}^{M-2}(\cdots(\mathrm{trunc}^{\ell+1}(s))\cdots)), \qquad \ell = 0, \ldots, M-2.$$

The truncated hierarchical B-spline (THB-spline) basis of $S_{\mathcal{H}}$ is obtained by applying the cumulative truncation operator to the elements of $\mathcal{H}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}})$ (Giannelli et al., 2012).

Definition 2. The truncated hierarchical B-spline basis $\mathcal{T}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}})$ of degree $\mathbf{d}$ with respect to the mesh $\mathcal{G}_{\mathcal{H}}$ is defined as

$$\mathcal{T}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}}) := \left\{ T_J^{\ell} \,:\, J \in A_{\mathbf{d}}^{\ell},\ \ell = 0, \ldots, M-1 \right\} \quad \text{with} \quad T_J^{\ell} := \mathrm{Trunc}^{\ell+1}(B_J^{\ell}), \quad \ell = 0, \ldots, M-2, \quad T_J^{M-1} = B_J^{M-1},$$

and B_J^{ℓ} is called the *mother* B-spline of T_J^{ℓ}.

THB-splines form a *convex partition of unity* and satisfy the *preservation of coefficients* property. This second property implies that truncated splines preserve the coefficients of functions represented with respect to any of the bases $\mathcal{B}_{\mathbf{d}}^{\ell}$. They are also a strongly stable basis for $S_{\mathcal{H}}$ with respect to the supremum norm (Giannelli et al., 2014), unlike the classical hierarchical basis of Definition 1, which is only weakly stable (Kraft, 1998).

2.2. Two-stage scattered data approximation scheme

Let

$$F = \{(\mathbf{X}_i, \mathbf{f}_i),\ i = 1, \ldots, n,\ \mathbf{X}_i \in \Omega \subset \mathbf{R}^2,\ \mathbf{f}_i \in \mathbf{R}^3, \text{with } \mathbf{X}_i \neq \mathbf{X}_j \text{ if } i \neq j\}, \tag{1}$$

be a scattered data set. We want to construct a THB-spline approximation of the data in F, by exploiting the local refinement capabilities of this adaptive spline construction, while simultaneously taking into account the non-uniform data distribution. In order to do this, we will combine suitable local polynomial approximations of variable degree with hierarchical quasi-interpolation.

2.2.1. Hierarchical quasi-interpolation

One of the advantages of considering the truncated basis for hierarchical spline spaces relies in the possibility of easily defining a hierarchical quasi-interpolation scheme without additional efforts with respect to its tensor-product counterpart.

We define the hierarchical quasi-interpolant (QI) $Q_{\mathcal{H}}$ in terms of the truncated basis as

$$Q_{\mathcal{H}}(F) := \sum_{\ell=0}^{M-1} \sum_{J \in A_{\mathbf{d}}^{\ell}} \boldsymbol{\lambda}_J^{\ell}(F)\, T_J^{\ell}, \tag{2}$$

where at any level each $\boldsymbol{\lambda}_J^{\ell}(F) \in \mathbb{R}^3$ coincides with the coefficient in the tensor-product quasi-interpolation scheme of level ℓ (Speleers and Manni, 2016). This means that the THB-spline property of *preservation of coefficients* enables the construction of quasi-interpolation operators that do not require additional computations with respect to the tensor-product case. More precisely, for any tensor-product spline space V^{ℓ} spanned by the B-spline basis $\mathcal{B}_{\mathbf{d}}^{\ell} := \{B_J^{\ell}, J \in \Gamma_{\mathbf{d}}^{\ell}\}$, let Q^{ℓ} be a spline quasi-interpolation operator of the form

$$Q^{\ell}(F) := \sum_{J \in \Gamma_{\mathbf{d}}^{\ell}} \boldsymbol{\lambda}_J^{\ell}(F)\, B_J^{\ell}. \tag{3}$$

The coefficient $\boldsymbol{\lambda}_J^{\ell}(F)$ associated to the truncated basis function T_J^{ℓ} in (2) is the one of its mother function B_J^{ℓ} in (3). Note that, if the data set F is obtained by sampling (for each component) a polynomial of degree $\mathbf{d}$, $Q_{\mathcal{H}}$ exactly reproduces it whenever any Q^{ℓ}, for $\ell = 0, \ldots, M-1$, has this property. Additional results on hierarchical quasi-interpolation with THB-splines include Bracco et al. (2016), Speleers (2017).

2.2.2. First stage: local variable-degree polynomial approximations

The first step to construct our adaptive approximation scheme requires the definition of the vector coefficients in (2). For each $\boldsymbol{\lambda}_J^{\ell}(F) = (\lambda_{J,1}^{\ell}(F), \lambda_{J,2}^{\ell}(F), \lambda_{J,3}^{\ell}(F))$, we consider a local subset $F_J^{\ell} \subseteq F$ defined as

$$F_J^\ell := \{(\mathbf{X}_i, \mathbf{f}_i) : i \in I_J^\ell\},$$

with

$$I_J^\ell := \left\{1 \le i \le n : \|\mathbf{X}_i - C_J^\ell\| \le \rho_J^\ell\right\}, \qquad \rho_J^\ell := \frac{\mathrm{diam}(\mathrm{supp}\, B_J^\ell)}{2},$$

where C_J^ℓ denotes the center of $\mathrm{supp}\, B_J^\ell$. If F_J^ℓ does not contain any element, ρ_J^ℓ is increased by a factor k. In order not to lose the locality of the method, we set a maximum enlarging factor K_J^ℓ such that $k \le K_J^\ell$.

Then, we construct a polynomial least squares fitting for the data set in F_J^ℓ. In order to do this, we consider the space Π_J^ℓ of polynomials of total degree d_J^ℓ generated by the basis $\mathcal{P}_J^\ell := \{p_1, \ldots, p_{\dim(\Pi_J^\ell)}\}$. The degree d_J^ℓ is initialized as the maximum value less than or equal to $d := \min_{h=1,2} d_h$ such that: (a) $\dim(\Pi_J^\ell) \le |F_J^\ell|$, and (b) the matrix

$$A_J^\ell := [p(X_i)]_{p \in \mathcal{P}_J^\ell, i \in I_J^\ell},$$

of the least square problem has full rank. With this assumption on A_J^ℓ, the vector coefficients of the least squares approximation in the local polynomial basis $\mathcal{P}_j^\ell$ are given by

$$((A_J^\ell)^T A_J^\ell)^{-1} (A_J^\ell)^T D_J^\ell, \qquad (D_J^\ell)_{h,k} := f_{j(h),k}, \qquad h = 1, \ldots, |F_J^\ell|, \qquad k = 1, 2, 3,$$

with $j : \{1, 2, \ldots, |F_J^\ell|\} \to I_J^\ell$ being a bijective map and $\mathbf{f}_i = (f_{i,1}, f_{i,2}, f_{i,3})$, $i = 1, .., n$. The value of $\boldsymbol{\lambda}_J^\ell(f)$ is defined as the vector coefficient associated with B_J^ℓ in the representation of the polynomial in the local spline space V_J^ℓ spanned by the B-splines $B_I^\ell, I \in \Gamma_{\mathbf{d}}^\ell$, which are active on $\mathrm{supp}\, B_J^\ell$. More explicitly, $\boldsymbol{\lambda}_J^\ell(f)$ is the row corresponding to B_J^ℓ in

$$\alpha_J^\ell := G_J^\ell D_J^\ell \in \mathbb{R}^{\dim(V_J^\ell) \times 3}, \qquad \text{with } G_J^\ell := (E_J^\ell)^T ((A_J^\ell)^T A_J^\ell)^{-1} (A_J^\ell)^T,$$

where E_J^ℓ is the matrix necessary to represent the polynomials of Π_J^ℓ in the B-spline basis of V_J^ℓ. Since the 2-norm of the matrix G_J will be an ingredient in the error estimates for the quasi-interpolant, we also check its value in order to be able to control it, when necessary. In other words, if it is not below a given threshold σ, we further decrease the degree of the local polynomial space and solve the least squares problem in the updated space.

Remark 1. Note that the matrix G_J^ℓ does not depend on the choice of the basis $\{p_1, \ldots, p_{\dim(\Pi_J^\ell)}\}$ for the polynomial space Π_J^ℓ. Let us consider another basis $p'_1, \ldots, p'_{\dim(\Pi_J^\ell)}$, and let

$$[p'_1(\cdot), \ldots, p'_{\dim(\Pi_J^\ell)}(\cdot)]^T = W\, [p_1(\cdot), \ldots, p_{\dim(\Pi_J^\ell)}(\cdot)]^T,$$

where $W \in \mathbb{R}^{\dim(\Pi_J^\ell) \times \dim(\Pi_J^\ell)}$ is the non-singular matrix representing the change of basis. If we denote by $A_J^\ell, A_J^{\ell\prime}$ and $E_J^\ell, E_J^{\ell\prime}$ the corresponding collocation matrices and matrices to represent the polynomials in the local B-spline basis, we have $A_J^{\ell\prime} = A_J^\ell W^T$ and $E_J^{\ell\prime} = W E_J^\ell$. As a consequence,

$$\begin{aligned} G_J^{\ell\prime} &= (E_J^{\ell\prime})^T ((A_J^{\ell\prime})^T A_J^{\ell\prime})^{-1} (A_J^{\ell\prime})^T = (E_J^\ell)^T W^T (W (A_J^\ell)^T A_J^\ell W^T)^{-1} W (A_J^\ell)^T \\ &= (E_J^\ell)^T W^T W^{-T} (A_J^\ell)^{-1} (A_J^\ell)^{-T} W^{-1} W (A_J^\ell)^T = (E_J^\ell)^T ((A_J^\ell)^T A_J^\ell)^{-1} (A_J^\ell)^T = G_J^\ell, \end{aligned}$$

that is, G_J^ℓ does not depend on the choice of the polynomial basis.

Remark 2. If we denote by $(\alpha_{J,1}^\ell, \alpha_{J,2}^\ell, \alpha_{J,3}^\ell)$ and $(D_{J,1}^\ell, D_{J,2}^\ell, D_{J,2}^\ell)$ the columns of α_J^ℓ and D_J^ℓ respectively, it is straightforward to obtain

$$|\lambda_{J,k}^\ell(F)| \le \|\alpha_{J,k}^\ell\|_2 \le \|G_J^\ell\|_2 \, \|D_{J,k}^\ell\|_2\,, \qquad k = 1, 2, 3.$$

Note that $\|G_J^\ell\|_2$ depends on the degree of the local polynomial approximation and on the number and distribution of the local set of data points $\{\mathbf{X}_i : i \in I_J^\ell\}$.

2.2.3. Second stage: adaptive scheme

The second stage of our fitting method defines the adaptive scheme by specifying the construction of the hierarchical space $S_{\mathcal{H}}$.

Initially, $S_{\mathcal{H}}$ coincides with a tensor product space V^0, and the hierarchical mesh $\mathcal{G}_{\mathcal{H}}$ with a uniform mesh $\mathcal{G}^0$. At each iteration the hierarchical QI in (3) with vector coefficients $\boldsymbol{\lambda}_J^\ell$, $J \in \Gamma^\ell$, $\ell = 0, \ldots, M-1$, is computed according to the local variable-degree polynomial approximation. The maximum enlarging factor K_J^ℓ used to determine the local data F_J^ℓ is

$$K_J^\ell = \left(\left\lceil \frac{2\delta^{\ell-1}}{\rho_J^\ell} \right\rceil + 1 \right), \tag{4}$$

where $\delta^\ell := \max_{J \in \Gamma_{\mathbf{d}}^\ell} \rho_J^\ell$. For K_J^0 we need to define δ^{-1} and ρ_J^{-1}: this is achieved by simply considering a B-spline set $\mathcal{B}_{\mathbf{d}}^{-1} := \{B_J^{-1},\ J \in \Gamma_{\mathbf{d}}^{-1}\}$ defined on an auxiliary tensor-product mesh whose size is twice the size of $\mathcal{G}^0$ and having at least $(d_1+1)(d_2+1)$ elements. Since our experience suggests that data points too far from the mesh elements that belong to the support of a certain THB-spline of level ℓ usually worsen the overall accuracy of the quasi-interpolant, the value of K_J^ℓ is chosen by taking into account the mesh size of the level. In addition, this enlargement factor also ensures, in combination with the refinement strategy, that enough data points are locally available to properly compute the coefficients associated with newly inserted basis functions, see also Remark 2 in Bracco et al. (2017a).

Subsequently, $Q_{\mathcal{H}}(F)$ is evaluated at all data points to get the errors

$$e_i := \| Q_{\mathcal{H}}(F)(\mathbf{X}_i) - \mathbf{f}_i \|_2, \qquad i = 1, \ldots, n.$$

The refinement of the hierarchical mesh is done in the areas of the domain where these errors exceed a given tolerance ϵ. More precisely, we perform dyadic refinement (in the two parametric directions) in the cells which belong to the support of the B-splines B_J^ℓ, $J \in \Gamma^\ell$, $\ell = 0, \ldots, M-1$ that contains at least 1 point $\mathbf{X}_i$ where $e_i > \epsilon$, and at least n_{loc} data points. A function-based refinement strategy of this kind is a natural choice in the design of any approximation scheme based on adaptive splines to guarantee the construction of a suitably refined spline space. Moreover, in view of our choice for K_J^ℓ, this refinement by functions, and in particular the parameter n_{loc}, influences the minimum number of local data used to compute the coefficients of the new basis functions in the first stage of the method. High values of this parameter generally lead to a less refined mesh, and, consequently, to a reduced number of degrees of freedom. Its value can be then chosen high enough to reduce as much as possible oscillations deriving from overfitting. An optimal compromise looks for the maximum value of n_{loc} which allows to have enough degrees of freedom to satisfy the given tolerance. Once the hierarchical mesh is refined, the hierarchical space and the number M of current levels are updated accordingly.

The whole procedure is repeated until the tolerance is satisfied for any point or, as in common industrial applications where the fitting of scanned data is employed, until it is satisfied for a certain percentage of the data points. If the algorithm does not provide the required tolerance before reaching a given maximum number of levels M_{max}, the overall procedure stops. The two-stage approximation scheme is summarized in Fig. 3.

3. Error analysis

By assuming that the data set F is obtained through the sampling of a suitably smooth function $f : \Omega \to \mathbb{R}^3$, we now derive an estimate for the local approximation error of our scheme in any cell $c \in \mathcal{G}_{\mathcal{H}}$. Consequently, we denote the QI by $Q_{\mathcal{H}}(f) := Q_{\mathcal{H}}(F)$. In order to simplify the notation, the analysis is developed for the scalar case, that is assuming $f : \Omega \to \mathbb{R}$. Remark 4 addresses the straightforward extension to the vector case.

In order to derive the error estimate, we need to consider *admissible meshes*, see Buffa and Giannelli (2016).

Definition 3. A hierarchical mesh $\mathcal{G}_{\mathcal{H}}$ with M levels is admissible of class m, $1 \le m < M$, if the THB-splines taking non-zero values on any cell $c \in \mathcal{G}_{\mathcal{H}}$ belong to at most m successive levels.

In view of the THB spline definition and the local support of B-splines, an easy consequence of assuming admissible meshes of a certain class m is that there exists a positive constant $\beta \ge 1$ depending on m such that $\forall c \in \mathcal{G}_{\mathcal{H}}$ we have

$$\operatorname{diam}(\operatorname{supp} B_J^\ell) \le \beta \operatorname{diam}(c)\,, \quad \forall J \in A^\ell(c), \quad \ell = 0, \ldots, M-1\,, \tag{5}$$

where

$$A^\ell(c) := \{J \in A_{\mathbf{d}}^\ell : c \subset \operatorname{supp} T_J^\ell\}\,. \tag{6}$$

Thus let c be any cell of $\mathcal{G}_{\mathcal{H}}$ and let us assume that $f \in C^{d(c)+1}(\mathcal{I}(c))$, where $\mathcal{I}(c)$ is the local enlargement of c,

$$\mathcal{I}(c) := \bigcup_{\ell=0}^{M-1} \bigcup_{J \in A^\ell(c)} \operatorname{supp} \lambda_J^\ell\,, \tag{7}$$

and where

$$d(c) := \min_{\ell=0,\ldots,M-1} \min_{J \in A^\ell(c)} d_J^\ell, \tag{8}$$

with $d_J^\ell \le d$ denoting the degree of the local polynomial approximation defined in the first stage of our method to compute the hierarchical QI coefficient associated with T_J^ℓ, for any $J \in A^\ell(c)\,, \ell = 0, \ldots, M-1$. For any polynomial p of degree less

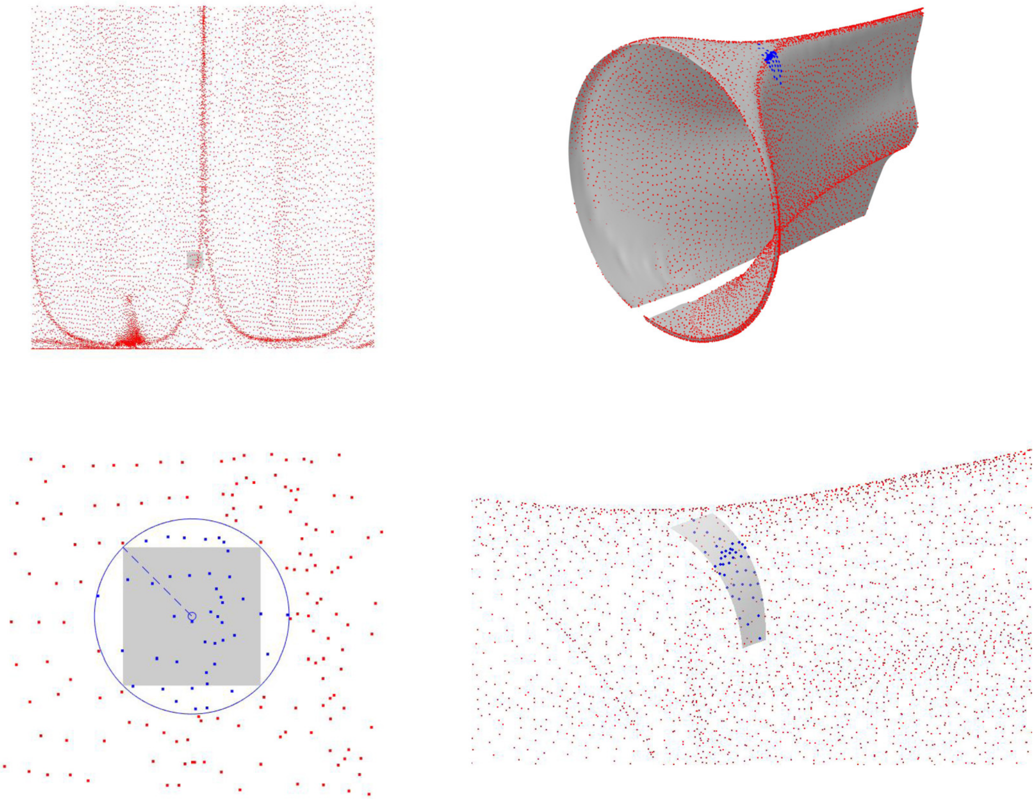


Fig. 3. Two-stage scattered data approximation scheme. Top left: the top view of the scattered data (red dots) is shown together with the support of a B-spline of a certain level (gray region). Bottom: the local approximation (gray region on the right plot) associated to a certain basis function (whose support is depicted in gray on the left plot) is computed by considering a local data set (blue dots) in the first stage of the method. The final approximation computed in the second stage of the method is also shown (top right). (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

or equal to $d(c)$, $Q_{\mathcal{H}}(p)|_c = p|_c$, since all local polynomial approximations involved in our scheme for defining $Q_{\mathcal{H}}(p)|_c$ coincide with p itself. Then, the following inequality holds:

$$\|f - Q_{\mathcal{H}}(f)\|_{\infty,c} \le \|f - p\|_{\infty,c} + \|Q_{\mathcal{H}}(f) - Q_{\mathcal{H}}(p)\|_{\infty,c} = \|f - p\|_{\infty,c} + \|Q_{\mathcal{H}}(f - p)\|_{\infty,c}\,. \tag{9}$$

Note that

$$\|Q_{\mathcal{H}}(f-p)\|_{\infty,c} = \left\| \sum_{\ell=0}^{M-1} \sum_{J\in A^\ell(c)} \lambda_J^\ell(f-p)\, T_J^\ell \right\|_{\infty,c} \le \max_{\substack{J\in A^\ell(c)\\ 0\le \ell < M}} |\lambda_J^\ell(f-p)|\,, \tag{10}$$

and that the first stage of our scheme implies

$$|\lambda_J^\ell(f-p)| \le \|G_J^\ell\|_2\, \|f-p\|_{\infty,\operatorname{supp}\lambda_J^\ell} \le \sigma\, \|f-p\|_{\infty,\operatorname{supp}\lambda_J^\ell}\,, \tag{11}$$

where σ is a given positive threshold. Since $c \subseteq \operatorname{supp}\lambda_J^\ell$, $\forall J \in A^\ell(c), \ell = 0,\dots,M-1$, from (9), (10) and (11), we obtain

$$\|f - Q_{\mathcal{H}}(f)\|_{\infty,c} \le (1+\sigma) \max_{\substack{J\in A^\ell(c)\\ 0\le \ell < M}} \|f-p\|_{\infty,\operatorname{supp}\lambda_J^\ell}\,. \tag{12}$$

In particular, if p is the truncated Taylor expansion of degree $d(c)$ of f developed at the center (x_c, y_c) of c, that is

$$p(x,y) = \sum_{r+s\le d(c)} \frac{\partial f^{r+s}(x_c,y_c)}{\partial x^r \partial y^s} \frac{(x-x_c)^r (y-y_c)^s}{r!s!}\,, \tag{13}$$

considering the Lagrange representation of its remainder for a sufficiently smooth function f, the following lemma can be proved.

Lemma 3. *For any $c \in \mathcal{G}_{\mathcal{H}}$, if $f \in C^{d(c)+1}(\mathcal{I}(c))$ with $d(c)$ and $\mathcal{I}(c)$ respectively defined in (8) and (7), for the Taylor expansion p introduced in (13) the following inequality holds true:*

$$\max_{\substack{J \in A^\ell(c) \\ 0 \le \ell < M}} \|f - p\|_{\infty, \mathrm{supp}\,\lambda_J^\ell} \le \left(\sum_{r+s=d(c)+1} \left\| \frac{\partial f^{r+s}}{\partial x^r \partial y^s} \right\|_{\infty, \mathcal{I}(c)} \frac{1}{r!s!} \right) \left(\max_{\substack{J \in A^\ell(c) \\ 0 \le \ell < M}} K_J^\ell \,\mathrm{diam}\left(\mathrm{supp}\, B_J^\ell\right) \right)^{d(c)+1}, \tag{14}$$

where $A^\ell(c)$ is defined in (6) and K_J^ℓ is the enlarging factor introduced in (4).

Proof. Since p is a polynomial of degree $d(c)$ the inequality in (12) holds true. Considering the Lagrange expression of the remainder, we then obtain

$$\max_{\substack{J \in A^\ell(c) \\ 0 \le \ell < M}} \|f - p\|_{\infty, \mathrm{supp}\,\lambda_J^\ell} \le \sum_{r+s=d(c)+1} \left\| \frac{\partial f^{r+s}}{\partial x^r \partial y^s} \right\|_{\infty, \mathcal{I}(c)} \frac{1}{r!s!} \left(\max_{J \in A^\ell(c),\, 0 \le \ell < M} \mathrm{diam}\left(\mathrm{supp}\,\lambda_J^\ell\right) \right)^{d(c)+1}.$$

The proof is completed by observing that the term in brackets on the right-hand side of the above equation does not depend on r and s and recalling that

$$\mathrm{diam}\left(\mathrm{supp}\,\lambda_J^\ell\right) \le K_J^\ell \,\mathrm{diam}\left(\mathrm{supp}\, B_J^\ell\right). \quad \square$$

Note that, since we are assuming that at each level the mesh $\Gamma_{\mathbf{d}}^\ell$ is uniform and obtained by a dyadic refinement of $\Gamma_{\mathbf{d}}^{\ell-1}$, we can state that

$$K_J^\ell \le 4(d+1) =: K\,. \tag{15}$$

Thus we can say that the right-hand side in (14) mainly depends on two terms: one related to the infinity norm in $\mathcal{I}(c)$ of all the derivatives of order $d(c)+1$ of the approximated function, and the other depending on the size of the supports of the B-splines of any level active on the considered cell. Under the assumed hypothesis of dealing with an admissible hierarchical mesh of class m, the size of these supports is related to the size of c as in (5). As a consequence, by also taking into account (15), the following corollary immediately descends from Lemma 3.

Corollary 1. *Given an admissible mesh $\mathcal{G}_{\mathcal{H}}$ of class m, $\forall c \in \mathcal{G}_{\mathcal{H}}$, if $f \in C^{d(c)+1}(\mathcal{I}(c))$, for the Taylor expansion p introduced in (13) the following inequality holds true,*

$$\max_{J \in A^\ell(c),\, 0 \le \ell < M} \|f - p\|_{\infty, \mathrm{supp}\,\lambda_J} \le (K\beta)^{d+1} \left(\sum_{r+s=d(c)+1} \left\| \frac{\partial f^{r+s}}{\partial x^r \partial y^s} \right\|_{\infty, \mathcal{I}(c)} \frac{1}{r!s!} \right) \left(\mathrm{diam}(c)\right)^{d(c)+1}, \tag{16}$$

where $d(c)$, β and K are respectively defined in (8), (5), and (15).

Finally, by combining Corollary 1 with the inequality in (12), we can derive the following main result.

Theorem 1. *Given an admissible mesh $\mathcal{G}_{\mathcal{H}}$ of class m, if $f \in C^{d+1}(\Omega)$, then $\forall c \in \mathcal{G}_{\mathcal{H}}$ the following inequality holds true,*

$$\|f - Q_{\mathcal{H}}(f)\|_{\infty, c} \le (1+\sigma)\,(K\beta)^{d+1} \left(\sum_{r+s=d(c)+1} \left\| \frac{\partial f^{r+s}}{\partial x^r \partial y^s} \right\|_{\infty, \mathcal{I}(c)} \frac{1}{r!s!} \right) \left(\mathrm{diam}(c)\right)^{d(c)+1}, \tag{17}$$

where $d(c)$, β and K are defined in (8), (5), and (15), respectively, while σ is the assigned positive threshold used in the first stage of the scheme.

Note that, if f is a smooth function, in order to obtain a fast convergence, this inequality suggests to use (when possible) the maximal value d for the degree $d(c)$. On the other hand, the local polynomial degree d_J^ℓ is chosen as the maximal non-negative integer less or equal to d guaranteeing that, once fixed the local sample F_J^ℓ, its size is greater than the dimension of the adopted local polynomial space Π_J^ℓ and also that $\|G_J^\ell\|_2 \le \sigma$. Thus, for some cell c, $d(c)$ can be lower than d or even equal to zero. This possibility cannot be avoided, since the local polynomial degree adaptivity is fundamental to ensure sufficiently robustness to our scheme, specially for highly varying data distributions. Obviously, in cases where the variable degree nature of the local least square approximation is not necessary and the local polynomials are always of maximal degree d—namely, $d(c) = d$ for any cell of the hierarchical mesh—the estimate derived in Theorem 1 implies a convergence

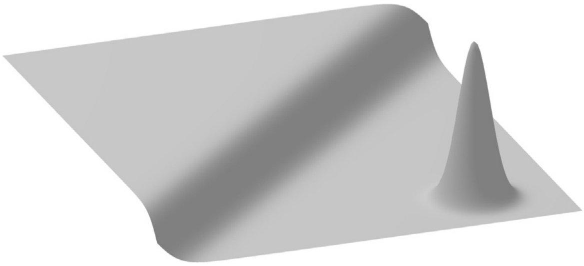
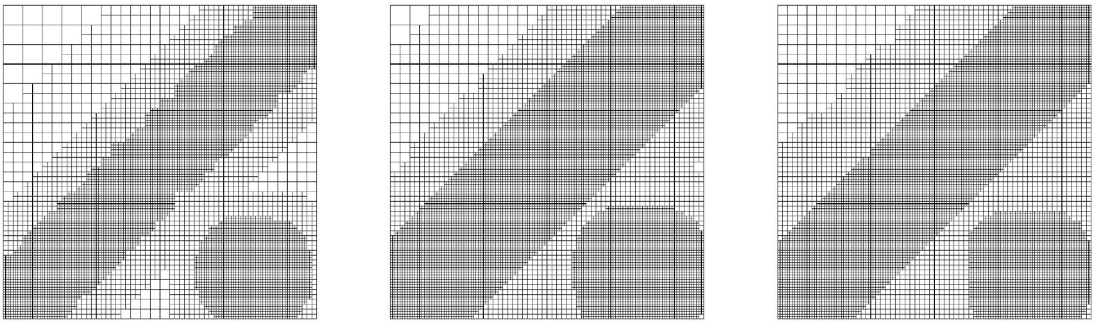


Fig. 4. Test surface of Example 1.



(a) $\mathbf{d} = (2, 2)$ (b) $\mathbf{d} = (3, 3)$ (c) $\mathbf{d} = (4, 4)$

Fig. 5. Hierarchical meshes of Example 1.

rate of h^{d+1}, where h denotes the mesh size of the finest level $M - 1$. In practice, when the local approximations are almost always of maximal degree d, this optimal behavior can still be preserved, as shown in the following example.

Example 1. We consider 3 sets of $n = 30000, 70000, 120000$ uniformly scattered data points, respectively. The data points X_i, $i = 1, ..., n$, are Halton points (generated with the MATLAB function `haltonset`) and the values f_i were obtained by evaluating the test function

$$f(x, y) = (\tanh(9y - 9x) + 1)/9 + (1.5\exp((10x - 6)^2 + (10y + 7)^2))^{-1}, \qquad (x, y) \in \Omega = [-1, 1]^2, \tag{18}$$

shown in Fig. 4.

The size of the considered sets of scattered data was chosen in order to have a density of the data sufficient to have $d(c) = d$ for almost all cells c. In this case, it is then possible to achieve the maximum order of convergence. For all the tests, we used $\epsilon = 10^{-4}$, $n_{loc} = 3$, $\sigma = 10^8$ and $M_{max} = 6$. The hierarchical meshes obtained for $\mathbf{d} = (2, 2), (3, 3), (4, 4)$ are reported in Fig. 5. As expected, we get more refinement where the test function has sharp variations. This means that in these regions of the domain the norm of the derivatives is larger, and, consequently, a smaller $\mathrm{diam}(c)$ is needed to have low error values, according to (17). Let us define

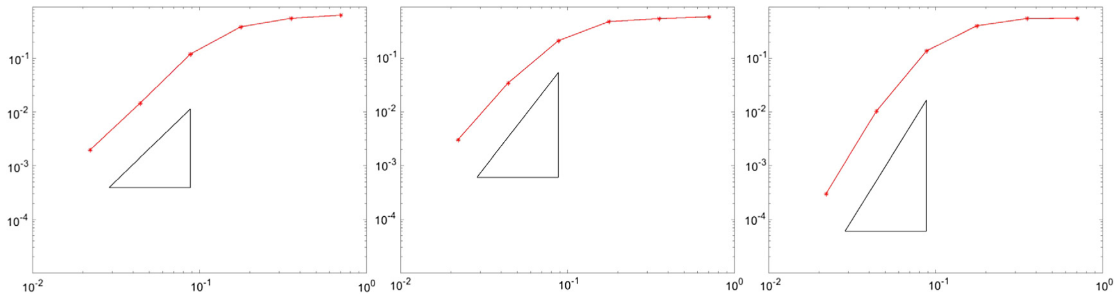
$$e_F := \max_{1 \le i \le n} \| Q_{\mathcal{H}}(f)(\mathbf{X}_i) - f(\mathbf{X}_i) \|_2,$$

and

$$e_\infty := \max_{\bar{\mathbf{X}} \in \bar{F}} \| Q_{\mathcal{H}}(f)(\bar{\mathbf{X}}) - f(\bar{\mathbf{X}}) \|_2, \qquad \bar{F} := \left\{ \left(-1 + 2\frac{i}{300}, -1 + 2\frac{j}{300} \right) : 0 \le i, j \le 300 \right\}.$$

Since we are able to have almost always the maximum degree for the local approximations, both e_F and e_∞ follow the theoretical convergence order h^{d+1}. Fig. 6 shows the behavior of e_∞, almost identical to e_F. Note that both errors coincide with those obtained using the tensor-product version of the quasi-interpolant, namely by assuming global refinement at each refinement step. This means that we are able to get the same accuracy using significantly less degrees of freedom.

Remark 4. When $f = (f_1, f_2, f_3)$, with $f_j : \Omega \to \mathbb{R}$, $j = 1, \ldots, 3$, the hierarchical mesh produced is adaptively obtained by taking into account the Euclidean norm of the errors at each data site. The choice of the degree d_J^ℓ of the local polynomial used to define the coefficient associated with T_J^ℓ on the final hierarchical mesh depends only on the local data points $\mathbf{X}_i$ in



(a) $\mathbf{d} = (2,2)$ (b) $\mathbf{d} = (3,3)$ (c) $\mathbf{d} = (4,4)$

Fig. 6. Convergence plots showing the behavior of e_∞ with respect to h for the approximations of the surface (18) in Example 1. The triangles represent the theoretical order of convergence.

F_J^ℓ, and therefore is not different. Consequently, the above error analysis can be applied for each component and, under the assumptions of Theorem 1, we can easily derive that

$$\sup_{\mathbf{X}\in c} \| f(\mathbf{X}) - Q_{\mathcal{H}}(f)(\mathbf{X})\|_2 \le (1+\sigma)\,(K\beta)^{d+1} \max_{j=1,\ldots,3} \left(\sum_{r+s=d(c)+1} \left\| \frac{\partial f_j^{r+s}}{\partial x^r \partial y^s} \right\|_{\infty,\mathcal{I}(c)} \frac{1}{r!s!} \right) \Big(\operatorname{diam}(c)\Big)^{d(c)+1}.$$

4. Reconstruction of aircraft engine parts

In this section we test the adaptive scattered data fitting scheme on two data sets obtained from the scan of the aircraft engine parts shown in Fig. 2 and parametrized by the method introduced in Floater (1997). Both parts are very challenging for a fully-automatic surface reconstruction with respect to fitting accuracy and stability because of their shapes defined by very high and very low curvature regions. This also results in scanned data with highly varying density. The typically required fitting errors for the parts of interest range from 10^{-4} m to 10^{-5} m for about 90% to 95% of the points.

As it was shown in Kiss et al. (2014), the use of adaptive splines is the key to generate smooth surfaces in a very efficient manner. For this reason, this new technology was recently brought into industrial environments. In particular, the significant reduction of the overall number of degrees of freedom while increasing the fitting accuracy in the high curvature regions, like the leading and trailing edge of blades, is a competitive advantage and great improvement of the currently available CAD standard. The global adaptive fitting algorithm based on regularized THB-spline least squares is defined as

$$s_{\mathcal{H}} := \sum_{\ell=0}^{M-1} \sum_{J\in A_{\mathbf{d}}^\ell} \mathbf{c}_J^\ell \, T_J^\ell, \tag{19}$$

such that

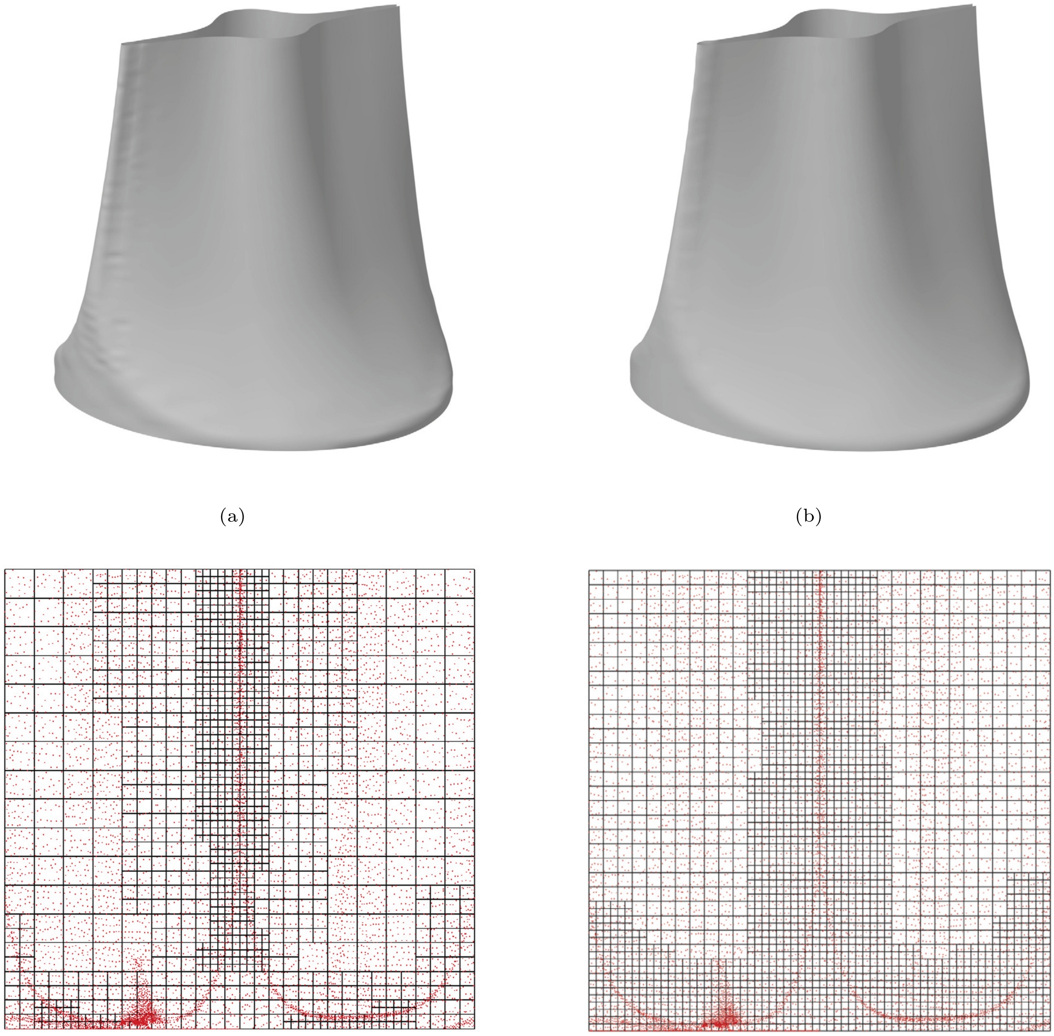
$$\sum_{i=1}^{n} \|s_{\mathcal{H}}(\mathbf{X}_i) - \mathbf{f}_i\|_2^2 = \min_{s\in S_{\mathcal{H}}} \sum_{i=1}^{n} \|s(\mathbf{X}_i) - \mathbf{f}_i\|_2^2 + \lambda J(s), \tag{20}$$

where λ is the smoothing coefficient and $J(s)$ is the thin-plate energy

$$J(s) := \int_\Omega \left\| \frac{\partial^2 s}{\partial x \partial x} \right\|_2^2 + 2\left\| \frac{\partial^2 s}{\partial x \partial y} \right\|_2^2 + \left\| \frac{\partial^2 s}{\partial y \partial y} \right\|_2^2 dx\, dy.$$

The energy term is used to avoid singularities in the linear system (which has to be solved during the fitting procedure) because of too few data in some areas and also to smooth the resulting surface. The hierarchical mesh (and the corresponding space) are obtained by refining the neighborhood of the cells containing the points $\mathbf{X}_i$ where the error is above a certain given tolerance ϵ, see Kiss et al. (2014) for details. This iterative procedure stops when the error is below the tolerance in at least a certain percentage of the data points $\mathbf{X}_i$, $i = 1,\ldots,n$.

This global fitting approach does not allow to take into account the local properties of the data, especially the data density and distribution. This may lead to unwanted oscillations in regions where too few data are available with respect to the considered refinement. As a consequence, the surface reconstruction step, as a part of the automatized multi-disciplinary reverse engineering process, tends to produce surfaces with some areas affected by oscillations. The following two examples will illustrate these effects and benchmark the new local scheme.



(a) (b) (c) (d)

Fig. 7. Results of Example 2: the reconstructed surfaces obtained with global least squares fitting (a) and with the new local scheme (b). The corresponding hierarchical meshes are also shown (c)–(d).

Example 2. In this example, we use the fitting algorithm to reconstruct a part of the tensile from the set of 9281 scattered data shown in Fig. 2(b). We ran both algorithms with $\mathbf{d} = (2, 2)$, $\epsilon = 5 \cdot 10^{-5}$ m, and starting with a 4×4 tensor-product mesh $\mathcal{G}^0$.

We applied the adaptive global least squares fitting, with smoothing coefficient $\lambda = 10^{-6}$ and 4 levels, obtaining 1136 degrees of freedom, 99.63% of points below the tolerance, and a maximum error of $8.33298 \cdot 10^{-5}$ m. The obtained reconstructed surface and hierarchical mesh are shown in Fig. 7(a)–(c). Note that the approximation is greatly affected by unwanted oscillations. This is a known phenomenon in this type of adaptive fitting algorithms, occurring in areas with heavy refinement but few available data with respect to the level of refinement. In fact, further tests done by using more refinement levels and higher values of the smoothing coefficient also failed to avoid the oscillations.

The local algorithm was used in this case with $n_{loc} = 20$, $\sigma = 10^8$ and 4 levels, and produced an approximation with 2506 degrees of freedom, that satisfies the same tolerance in 96.25% of the data points $\mathbf{X}_i$ with maximum error $1.22332 \cdot 10^{-4}$ m. The surface, free of oscillations, and the hierarchical mesh are shown in Fig. 7(b)–(d).

Example 3. In this example, we test the scheme on the set of 27191 scattered data shown in Fig. 2(a). We ran both the algorithms with $\mathbf{d} = (3, 3)$, $\epsilon = 2 \cdot 10^{-5}$ m, and starting with a 4×4 tensor-product mesh $\mathcal{G}^0$.

The adaptive global least squares fitting, with smoothing coefficient $\lambda = 10^{-8}$ and 6 levels, gives 8164 degrees of freedom, 99.99% of points below the tolerance and a maximum error of $2.11625 \cdot 10^{-5}$ m. The obtained reconstructed surface and hierarchical mesh are shown in Fig. 8(a)–(c). Note that, also in this case, the surface shows many oscillations, see Fig. 8(a).

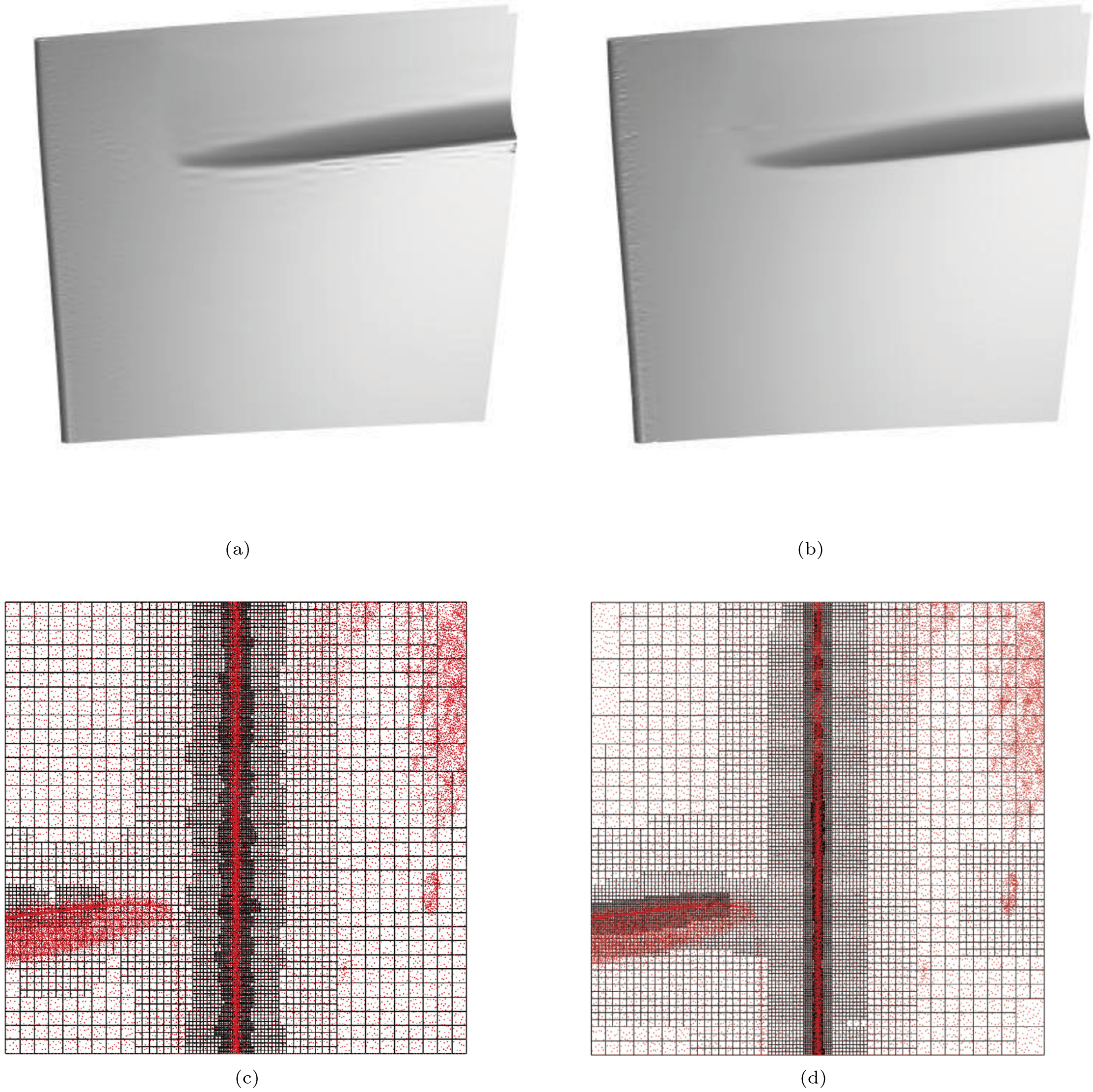


Fig. 8. Results of Example 3: the reconstructed surfaces obtained with global least squares fitting (a) and with the new local scheme (b). The corresponding hierarchical meshes are also shown (c)–(d).

Our local scheme, with minimum number of local data $n_{loc} = 60$, $\sigma = 10^8$ and 7 levels, produces an approximation using 12721 degrees of freedom, which satisfies the same tolerance in 97.02% of the data points with maximum error $4.88418 \cdot 10^{-5}$ m. The surface, which is based on the mesh in Fig. 8(d), in some areas presents a reduction of the oscillations obtained with the other method, as can be seen from Fig. 8(b).

In the numerical tests we could show that the newly introduced fitting procedure can generate high-quality surfaces with respect to accuracy and stability out of scanned data of industrial complexity. Furthermore, the locality of the scheme and its ability to control the number n_{loc} of locally available data often allows to reduce the oscillations which may appear with the global regularized least squares approximation. This feature allows to select the areas to be refined according not only to the local error, but also to the number of local data, which should be large enough to justify a further refinement. Of course n_{loc} should be always carefully chosen large enough to reduce oscillations, but not too large to be still able to sufficiently refine and get error values below the prescribed tolerance. The selection of a suitable choice for this parameter however, is usually not a problem since it can be easily identified with few numerical experiments. Even if the local method does not necessarily decrease the number of degrees of freedom when compared to the global approach, the possibility of reducing oscillations increases the reliability of the fitting algorithm, as well as the quality of the final hierarchical spline model. In addition, the local nature of the first stage of the method can be naturally exploited to improve the efficiency of the fitting framework (e.g., for parallelization).

5. Conclusion

We presented an adaptive surface reconstruction scheme which exploits the solution of local linear systems for the construction of an accurate and efficient THB-spline approximation of a given scattered data set. The adaptivity nature of the scheme is twofold: *shape adaptivity* (thanks to the local refinement capabilities of THB-splines), and *data adaptivity* (thanks to the local polynomial approximations of variable degree). The presented algorithm is independent from the polynomial basis used to compute the local least square polynomial approximations. Moreover, our refinement strategy, which allows to consider a varying number of local data points, ensures a significant flexibility to the overall adaptive construction. The performance of the scheme are theoretically confirmed by the error analysis of the method, and experimentally validated on industrial components of critical geometric models.

From the application point of view, the introduced local scheme is important to reduce peaks and oscillations in the generated surfaces as the result of an overdetermined fitting problem. By naturally integrating the data adaptivity into the geometry reconstruction step, the scheme enables fully automatic reverse engineering processes without any user-interaction or part-specific templates for a wide range of mechanical components. This is a further improvement to the ability of generating highly accurate surfaces by using hierarchical spline constructions and increases the importance of bringing these new technologies into standard CAD systems.

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Appendix A. Notation

The following list summarizes the notation adopted in the paper.

- V^ℓ: tensor-product spline space of degree $\mathbf{d} := (d_1, d_2)$ and level ℓ, for $\ell = 0, \ldots, M-1$;
- $\mathcal{G}^\ell$: tensor-product grid of level ℓ, for $\ell = 0, \ldots, M-1$;
- $\mathcal{B}_{\mathbf{d}}^\ell$: B-spline basis of V^ℓ, for $\ell = 0, \ldots, M-1$;
- Ω^ℓ: subdomain (of a given domain Ω) of level ℓ, for $\ell = 0, \ldots, M$, with $\Omega^{\ell+1} \subseteq \Omega^\ell$, $\Omega^0 = \Omega$ and $\Omega_M = \emptyset$;
- $\mathcal{G}_{\mathcal{H}}$: hierarchical mesh;
- $\mathcal{H}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}})$: hierarchical B-spline basis of degree $\mathbf{d}$ with respect to the mesh $\mathcal{G}_{\mathcal{H}}$;
- $\mathcal{T}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}})$: truncated hierarchical B-spline basis of degree $\mathbf{d}$ with respect to the mesh $\mathcal{G}_{\mathcal{H}}$;
- $S_{\mathcal{H}}$: hierarchical spline space spanned by the basis functions in $\mathcal{T}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}})$ (or $\mathcal{H}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}})$, equivalently);
- F: set of n scattered data points;
- $Q^\ell(\cdot)$: quasi-interpolant of level ℓ defined in V^ℓ;
- $Q_{\mathcal{H}}(\cdot)$: hierarchical quasi-interpolant defined in $S_{\mathcal{H}}$;
- K_J^ℓ maximum enlarging factor that controls the local data sample used to construct the local least squares polynomial approximation of any THB-spline $T_J^\ell \in \mathcal{T}_{\mathbf{d}}(\mathcal{G})$;
- σ: positive threshold for selecting the local polynomial approximation used to determine the coefficient associated with T_J^ℓ, for any $T_J^\ell \in \mathcal{T}_{\mathbf{d}}(\mathcal{G}_{\mathcal{H}})$;
- n_{loc}: minimum number of local data required in the refinement strategy.

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